

Submitted by: Ashvin Bhatt

Week: 3

Topic: SQL & Excel for Data Analytics

1. Task 1 – Introduction to Databases & SELECT Statement

Objective

To understand the basic structure of a relational database and practice retrieving data using the SELECT statement.

Database Used: PostgreSQL

Table Used: Customers

Brief Explanation

The **SELECT** statement is used to retrieve data from one or more columns of a table.

SELECT * retrieves all columns and records, while specific column names can be used to retrieve only the required information. Column aliases using **AS** provide more meaningful or readable column names in the output.

SQL Query

Query 1 – Display all records

```
SELECT * FROM customers;
```

Query 2 – Select specific columns

```
SELECT customer_name, city FROM customers;
```

Query 3 – Use column aliases

```
SELECT customer_id AS "Customer ID",  
       customer_name AS "Customer Name",  
       city AS "Customer City"  
FROM customers;
```

Screenshot of Output

Figure 1.1 – Displaying all customer records

Data OutputMessagesNotifications

≡+

▼

▼

SQL

	customer_id [PK] integer	customer_name character varying (100)	city character varying (50)
1	1	Rahul Sharma	Delhi
2	2	Priya Mehta	Mumbai
3	3	Amit Patel	Ahmedabad
4	4	Neha Singh	Pune
5	5	Rohan Verma	Delhi

Total rows: 5Query complete 00:00:00.316

Figure 1.2 – Selecting specific columns

Data OutputMessagesNotifications

≡+

▼

▼

SQL

	customer_name character varying (100)	city character varying (50)
1	Rahul Sharma	Delhi
2	Priya Mehta	Mumbai
3	Amit Patel	Ahmedabad
4	Neha Singh	Pune
5	Rohan Verma	Delhi

Total rows: 5Query complete 00:00:00.185

Figure 1.3 – Using column aliases

Data Output				Messages	Notifications
≡+ 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈 SQL					
	Customer ID integer	Customer Name character varying (100)	Customer City character varying (50)		
1	1	Rahul Sharma	Delhi		
2	2	Priya Mehta	Mumbai		
3	3	Amit Patel	Ahmedabad		
4	4	Neha Singh	Pune		
5	5	Rohan Verma	Delhi		
Total rows: 5				Query complete 00:00:00.173	

2. Task 2 – WHERE, ORDER BY & Aggregate Functions

Objective

To filter, sort, and summarize product data using SQL conditions, ordering, and aggregate functions.

Database Used: PostgreSQL

Table Used: Products

Query 1 – Filtering and Sorting

```
SELECT product_id, product_name, category, price
FROM products WHERE category = 'Electronics'
AND price > 1000
ORDER BY price DESC;
```

Explanation:

The query filters products belonging to the Electronics category with a price greater than ₹1,000. The filtered records are sorted by price in descending order.

Output:

Data Output Messages Notifications				
Showing rows: 1 to 4				
	product_id [PK] integer	product_name character varying (100)	category character varying (50)	price numeric (10,2)
1	101	Laptop	Electronics	60000.00
2	106	Monitor	Electronics	18000.00
3	108	Headphones	Electronics	2500.00
4	104	Keyboard	Electronics	1500.00
Total rows: 4 Query complete 00:00:00.187				

Query 2 – Aggregate Functions

```
SELECT COUNT(*) AS total_products,  
       SUM(price) AS total_price,  
       ROUND(AVG(price), 2) AS average_price,  
       MIN(price) AS minimum_price,  
       MAX(price) AS maximum_price  
  
FROM products;
```

Explanation:

The query uses `COUNT()`, `SUM()`, `AVG()`, `MIN()`, and `MAX()` to calculate summary statistics for the product prices.

Output:

Data Output Messages Notifications						
Showing						
	total_products bigint	total_price numeric	average_price numeric	minimum_price numeric	maximum_price numeric	
1	8	111300.00	13912.50	800.00	60000.00	
Total rows: 1 Query complete 00:00:00.141						

3. Task 3 – GROUP BY & HAVING

Objective

To group product records by category and calculate summary statistics for each group using **GROUP BY**, aggregate functions, and **HAVING**.

Database Used: PostgreSQL

Table Used: Products

SQL Query:

```
SELECT category,  
       COUNT(*) AS total_products,  
       ROUND(AVG(price), 2) AS average_price  
FROM products  
GROUP BY category  
HAVING COUNT(*) > 2  
ORDER BY average_price DESC;
```

Explanation:

The **GROUP BY** clause groups products according to their category. **COUNT()** calculates the number of products in each category, while **AVG()** calculates the average product price. The **HAVING** clause filters the grouped results and retains only categories having more than two products. The final results are sorted by average price in descending order.

Output:

Data Output				Messages	Notifications
         SQL				Showing	
	category character varying (50) 	total_products bigint 	average_price numeric 		
1	Electronics	5	16560.00		
2	Furniture	3	9500.00		
Total rows: 2				Query complete 00:00:00.176	

4. Task 4 – SQL Joins

Objective

To combine related data from multiple tables using **INNER JOIN**, **LEFT JOIN**, and **RIGHT JOIN**.

Database Used: PostgreSQL

Table Used: customers, sales

Query 1 - INNER JOIN

SELECT

s.sale_id,

c.customer_name,

s.product_id,

```
s.quantity,  
  
s.sale_date
```

```
FROM sales s
```

```
INNER JOIN customers c
```

```
ON s.customer_id = c.customer_id;
```

Explanation:

INNER JOIN returns only records where a matching **customer_id** exists in both tables.

Output:

Data Output

Messages

Notifications

<

Query 2 - LEFT JOIN

```
SELECT
```

```
c.customer_id,  
  
c.customer_name,  
  
c.city,  
  
s.sale_id,  
  
s.quantity
```

```
FROM customers c
```

```
LEFT JOIN sales s
```

```
ON c.customer_id = s.customer_id
```

ORDER BY c.customer_id;

Explanation:

LEFT JOIN returns all records from the left table (**customers**) and matching records from the **sales** table.

Output:

Data Output

Messages

Notifications

≡

+

▼

▼

SQL

Showing rows: 1 to 8

	customer_id integer	customer_name character varying (100)	city character varying (50)	sale_id integer	quantity integer
1	1	Rahul Sharma	Delhi	1001	1
2	1	Rahul Sharma	Delhi	1006	1
3	2	Priya Mehta	Mumbai	1008	1
4	2	Priya Mehta	Mumbai	1002	2
5	3	Amit Patel	Ahmedabad	1007	2
6	3	Amit Patel	Ahmedabad	1003	1
7	4	Neha Singh	Pune	1004	3
8	5	Rohan Verma	Delhi	1005	2

Total rows: 8

Query complete 00:00:00.142

Query 3 - RIGHT JOIN

SELECT

c.customer_id,

c.customer_name,

s.sale_id,

s.product_id,

s.quantity

FROM customers c

RIGHT JOIN sales s

ON c.customer_id = s.customer_id

ORDER BY s.sale_id;

Explanation:

RIGHT JOIN returns all records from the right table (**sales**) and matching records from the **customers** table.

Output:

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 8

	customer_id integer	customer_name character varying (100)	sale_id integer	product_id integer	quantity integer
1	1	Rahul Sharma	1001	101	1
2	2	Priya Mehta	1002	102	2
3	3	Amit Patel	1003	103	1
4	4	Neha Singh	1004	104	3
5	5	Rohan Verma	1005	105	2
6	1	Rahul Sharma	1006	106	1
7	3	Amit Patel	1007	107	2
8	2	Priya Mehta	1008	108	1

Total rows: 8

Query complete 00:00:00.126

5. Task 5 – SQL Subqueries

Objective

To use subqueries to solve analytical problems involving aggregate calculations.

Database Used: PostgreSQL

Table Used: Products

Problem 1 – Products Above Average Price

```
SELECT *
```

```
FROM products
```

```
WHERE price > (
```

```
    SELECT AVG(price)
```

```
    FROM products
```

```
)
```

```
ORDER BY price DESC;
```

Explanation:

The inner query calculates the average product price. The outer query then returns products whose price is greater than this average.

Output:

Data Output Messages Notifications				
Showing rows: 1 to 2				
	product_id [PK] integer	product_name character varying (100)	category character varying (50)	price numeric (10,2)
1	101	Laptop	Electronics	60000.00
2	106	Monitor	Electronics	18000.00
Total rows: 2 Query complete 00:00:00.140				

Problem 2 – Highest-Priced Product

```
SELECT *  
FROM products  
WHERE price = ( SELECT MAX(price)  
                FROM products  
              );
```

Explanation:

The subquery finds the maximum product price, and the outer query retrieves the product having that price.

Output:

Data Output Messages Notifications				
Showing rows: 1 to 1				
	product_id [PK] integer	product_name character varying (100)	category character varying (50)	price numeric (10,2)
1	101	Laptop	Electronics	60000.00
Total rows: 1 Query complete 00:00:00.123				

[illegible]

Screenshot 6.2: Dataset filtered by Electronics category

	A	B	C	D	E	F	G	H	I	J	K
1	Sale	Sale_Dat	Customer_Nam	City	Product	Product_Nam	Categor	Quant	Unit_Pri	Payment_Mo	Salespers
2	1001	01-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	1	₹60,000.00	UPI	Ankit
3	1009	09-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	4	₹60,000.00	UPI	Neha
4	1017	02-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	2	₹60,000.00	UPI	Priya
5	1025	10-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	5	₹60,000.00	UPI	Vikram
6	1006	06-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	1	₹18,000.00	Credit Card	Ankit
7	1014	14-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	4	₹18,000.00	Credit Card	Neha
8	1022	07-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	2	₹18,000.00	Credit Card	Priya
9	1030	15-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	5	₹18,000.00	Credit Card	Vikram
21	1008	08-Sep-2026	Arjun Rao	Bengaluru	108	Headphones	Electronics	3	₹2,500.00	Cash	Rahul
22	1016	01-Sep-2026	Arjun Rao	Bengaluru	108	Headphones	Electronics	1	₹2,500.00	Cash	Ankit
23	1024	09-Sep-2026	Arjun Rao	Bengaluru	108	Headphones	Electronics	4	₹2,500.00	Cash	Neha
24	1004	04-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	4	₹1,500.00	Cash	Neha
25	1012	12-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	2	₹1,500.00	Cash	Priya
26	1020	05-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	5	₹1,500.00	Cash	Vikram
27	1028	13-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	3	₹1,500.00	Cash	Rahul
28	1002	02-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	2	₹800.00	Credit Card	Priya
29	1010	10-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	5	₹800.00	Credit Card	Vikram
30	1018	03-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	3	₹800.00	Credit Card	Rahul
31	1026	11-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	1	₹800.00	Credit Card	Ankit
32											

7. Task 7 – Conditional Formatting & Excel Functions

Objective

To use conditional formatting and Excel functions to analyze sales data and identify high-value sales transactions.

7.1 Conditional Formatting

Conditional formatting was applied to the **Sales_Value** column to highlight sales transactions where the sales value was **₹50,000 or more**.

This makes high-value transactions easy to identify visually.

7.2 IF Function

The **IF()** function was used to classify sales transactions into High and Normal categories.

=IF(L2>=50000,"High","Normal")

If the sales value is ₹50,000 or greater, the transaction is classified as **High**; otherwise, it is classified as **Normal**.

7.3 COUNTIF Function

The **COUNTIF()** function was used to count the number of high-value sales transactions.

=COUNTIF(M2:M31,"High")

The formula counts the number of cells in the **Sales_Level** column containing **"High"**.

7.4 SUMIF Function

The **SUMIF()** function was used to calculate the total sales value of Electronics products.

=SUMIF(G2:G31,"Electronics",L2:L31)

The formula adds the **Sales_Value** of all transactions where the product category is **Electronics**.

Screenshot:

O3																
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	
1	Sale_ID	Sale_Date	Customer_Name	City	Product_ID	Product_Name	Category	Quantity	Unit_Price	Payment_Mode	Salesperson	Sales_Value	Sales_Level	Metric	Result	
2	1001	01-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	1	₹60,000.00	UPI	Ankit	₹ 60,000.00	High	High Value Sales Count	7	
3	1009	09-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	4	₹60,000.00	UPI	Neha	₹ 2,40,000.00	High	Electronics Sales Value	₹ 9,85,800.00	
4	1017	02-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	2	₹60,000.00	UPI	Priya	₹ 1,20,000.00	High			
5	1025	10-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	5	₹60,000.00	UPI	Vikram	₹ 3,00,000.00	High			
6	1006	06-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	1	₹18,000.00	Credit Card	Ankit	₹ 18,000.00	Normal			
7	1014	14-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	4	₹18,000.00	Credit Card	Neha	₹ 72,000.00	High			
8	1022	07-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	2	₹18,000.00	Credit Card	Priya	₹ 36,000.00	Normal			
9	1030	15-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	5	₹18,000.00	Credit Card	Vikram	₹ 90,000.00	High			
10	1005	05-Sep-2026	Rohan Verma	Delhi	105	Desk	Furniture	5	₹12,000.00	UPI	Vikram	₹ 60,000.00	High			
11	1013	13-Sep-2026	Rohan Verma	Delhi	105	Desk	Furniture	3	₹12,000.00	UPI	Rahul	₹ 36,000.00	Normal			
12	1021	06-Sep-2026	Rohan Verma	Delhi	105	Desk	Furniture	1	₹12,000.00	UPI	Ankit	₹ 12,000.00	Normal			
13	1029	14-Sep-2026	Rohan Verma	Delhi	105	Desk	Furniture	4	₹12,000.00	UPI	Neha	₹ 48,000.00	Normal			
14	1007	07-Sep-2026	Simran Kaur	Chandigarh	107	Bookshelf	Furniture	2	₹9,000.00	Debit Card	Priya	₹ 18,000.00	Normal			
15	1015	15-Sep-2026	Simran Kaur	Chandigarh	107	Bookshelf	Furniture	5	₹9,000.00	Debit Card	Vikram	₹ 45,000.00	Normal			
16	1023	08-Sep-2026	Simran Kaur	Chandigarh	107	Bookshelf	Furniture	3	₹9,000.00	Debit Card	Rahul	₹ 27,000.00	Normal			
17	1003	03-Sep-2026	Amit Patel	Ahmedabad	103	Office Chair	Furniture	3	₹7,500.00	Debit Card	Rahul	₹ 22,500.00	Normal			
18	1011	11-Sep-2026	Amit Patel	Ahmedabad	103	Office Chair	Furniture	1	₹7,500.00	Debit Card	Ankit	₹ 7,500.00	Normal			
19	1019	04-Sep-2026	Amit Patel	Ahmedabad	103	Office Chair	Furniture	4	₹7,500.00	Debit Card	Neha	₹ 30,000.00	Normal			
20	1027	12-Sep-2026	Amit Patel	Ahmedabad	103	Office Chair	Furniture	2	₹7,500.00	Debit Card	Priya	₹ 15,000.00	Normal			
21	1008	08-Sep-2026	Arjun Rao	Bengaluru	108	Headphones	Electronics	3	₹2,500.00	Cash	Rahul	₹ 7,500.00	Normal			
22	1016	01-Sep-2026	Arjun Rao	Bengaluru	108	Headphones	Electronics	1	₹2,500.00	Cash	Ankit	₹ 2,500.00	Normal			
23	1024	09-Sep-2026	Arjun Rao	Bengaluru	108	Headphones	Electronics	4	₹2,500.00	Cash	Neha	₹ 10,000.00	Normal			
24	1004	04-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	4	₹1,500.00	Cash	Neha	₹ 6,000.00	Normal			
25	1012	12-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	2	₹1,500.00	Cash	Priya	₹ 3,000.00	Normal			
26	1020	05-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	5	₹1,500.00	Cash	Vikram	₹ 7,500.00	Normal			
27	1028	13-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	3	₹1,500.00	Cash	Rahul	₹ 4,500.00	Normal			
28	1002	02-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	2	₹800.00	Credit Card	Priya	₹ 1,600.00	Normal			
29	1010	10-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	5	₹800.00	Credit Card	Vikram	₹ 4,000.00	Normal			
30	1018	03-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	3	₹800.00	Credit Card	Rahul	₹ 2,400.00	Normal			
31	1026	11-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	1	₹800.00	Credit Card	Ankit	₹ 800.00	Normal			
32																

8. Task 8 – VLOOKUP & XLOOKUP

Objective

To retrieve product information from a related lookup table using **VLOOKUP()** and **XLOOKUP()**.

VLOOKUP

The **VLOOKUP()** function was used to search for a **Product_ID** in the **Product_Lookup** table and return the corresponding product name.

=VLOOKUP(E2,Product_Lookup!A2:D9,2,FALSE)

The formula searches for the Product ID in the first column of the lookup range and returns the corresponding product name from the second column.

XLOOKUP

The **XLOOKUP()** function was used to retrieve the product name using the same Product ID.

=XLOOKUP(E2,Product_Lookup!A2:A9,Product_Lookup!B2:B9)

XLOOKUP searches the Product ID in the specified lookup range and returns the corresponding value from the specified return range.

Difference Between VLOOKUP and XLOOKUP

VLOOKUP	XLOOKUP
Uses a column index number	Uses a separate return range
Lookup value must be in the first column of the selected range	Lookup and return ranges can be specified independently
Less flexible	More flexible
Older and widely supported	Newer and generally more versatile

Screenshot:

=XLOOKUP(E5,Product_Lookup!\$A\$2:\$A\$9,Product_Lookup!\$B\$2:\$B\$9)																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	Sale_ID	Sale_Date	Customer_Name	City	Product_ID	Product_Name	Category	Quantity	Unit_Price	Payment_Mode	Salesperson	Sales_Value	Sales_Level	Metric	Result	VLOOKUP_Result	XLOOKUP_Result
2	1001	01-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	1	₹60,000.00	UPI	Ankit	₹ 60,000.00	High	High Value Sales Count Electronics Sales Value	₹ 9,85,800.00	7 Laptop	Laptop
3	1009	09-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	4	₹60,000.00	UPI	Neha	₹ 2,40,000.00	High			Laptop	Laptop
4	1017	02-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	2	₹60,000.00	UPI	Priya	₹ 1,20,000.00	High			Laptop	Laptop
5	1025	10-Sep-2026	Rahul Sharma	Delhi	101	Laptop	Electronics	5	₹60,000.00	UPI	Vikram	₹ 3,00,000.00	High			Laptop	Laptop
6	1006	06-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	1	₹18,000.00	Credit Card	Ankit	₹ 18,000.00	Normal			Monitor	Monitor
7	1014	14-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	4	₹18,000.00	Credit Card	Neha	₹ 72,000.00	High			Monitor	Monitor
8	1022	07-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	2	₹18,000.00	Credit Card	Priya	₹ 36,000.00	Normal			Monitor	Monitor
9	1030	15-Sep-2026	Karan Malhotra	Jaipur	106	Monitor	Electronics	5	₹18,000.00	Credit Card	Vikram	₹ 90,000.00	High			Monitor	Monitor
10	1005	05-Sep-2026	Rohan Verma	Delhi	105	Desk	Furniture	5	₹12,000.00	UPI	Vikram	₹ 60,000.00	High			Desk	Desk
11	1013	13-Sep-2026	Rohan Verma	Delhi	105	Desk	Furniture	3	₹12,000.00	UPI	Rahul	₹ 36,000.00	Normal			Desk	Desk
12	1021	06-Sep-2026	Rohan Verma	Delhi	105	Desk	Furniture	1	₹12,000.00	UPI	Ankit	₹ 12,000.00	Normal			Desk	Desk
13	1029	14-Sep-2026	Rohan Verma	Delhi	105	Desk	Furniture	4	₹12,000.00	UPI	Neha	₹ 48,000.00	Normal			Desk	Desk
14	1007	07-Sep-2026	Simran Kaur	Chandigarh	107	Bookshelf	Furniture	2	₹9,000.00	Debit Card	Priya	₹ 18,000.00	Normal			Bookshelf	Bookshelf
15	1015	15-Sep-2026	Simran Kaur	Chandigarh	107	Bookshelf	Furniture	5	₹9,000.00	Debit Card	Vikram	₹ 45,000.00	Normal			Bookshelf	Bookshelf
16	1023	08-Sep-2026	Simran Kaur	Chandigarh	107	Bookshelf	Furniture	3	₹9,000.00	Debit Card	Rahul	₹ 27,000.00	Normal			Bookshelf	Bookshelf
17	1003	03-Sep-2026	Amit Patel	Ahmedabad	103	Office Chair	Furniture	3	₹7,500.00	Debit Card	Rahul	₹ 22,500.00	Normal			Office Chair	Office Chair
18	1011	11-Sep-2026	Amit Patel	Ahmedabad	103	Office Chair	Furniture	1	₹7,500.00	Debit Card	Ankit	₹ 7,500.00	Normal			Office Chair	Office Chair
19	1019	04-Sep-2026	Amit Patel	Ahmedabad	103	Office Chair	Furniture	4	₹7,500.00	Debit Card	Neha	₹ 30,000.00	Normal			Office Chair	Office Chair
20	1027	12-Sep-2026	Amit Patel	Ahmedabad	103	Office Chair	Furniture	2	₹7,500.00	Debit Card	Priya	₹ 15,000.00	Normal			Office Chair	Office Chair
21	1008	08-Sep-2026	Arjun Rao	Bengaluru	108	Headphones	Electronics	3	₹2,500.00	Cash	Rahul	₹ 7,500.00	Normal			Headphones	Headphones
22	1016	01-Sep-2026	Arjun Rao	Bengaluru	108	Headphones	Electronics	1	₹2,500.00	Cash	Ankit	₹ 2,500.00	Normal			Headphones	Headphones
23	1024	09-Sep-2026	Arjun Rao	Bengaluru	108	Headphones	Electronics	4	₹2,500.00	Cash	Neha	₹ 10,000.00	Normal			Headphones	Headphones
24	1004	04-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	4	₹1,500.00	Cash	Neha	₹ 6,000.00	Normal			Keyboard	Keyboard
25	1012	12-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	2	₹1,500.00	Cash	Priya	₹ 3,000.00	Normal			Keyboard	Keyboard
26	1020	05-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	5	₹1,500.00	Cash	Vikram	₹ 7,500.00	Normal			Keyboard	Keyboard
27	1028	13-Sep-2026	Neha Singh	Pune	104	Keyboard	Electronics	3	₹1,500.00	Cash	Rahul	₹ 4,500.00	Normal			Keyboard	Keyboard
28	1002	02-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	2	₹800.00	Credit Card	Priya	₹ 1,600.00	Normal			Mouse	Mouse
29	1010	10-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	5	₹800.00	Credit Card	Vikram	₹ 4,000.00	Normal			Mouse	Mouse
30	1018	08-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	3	₹800.00	Credit Card	Rahul	₹ 2,400.00	Normal			Mouse	Mouse
31	1026	11-Sep-2026	Priya Mehta	Mumbai	102	Mouse	Electronics	1	₹800.00	Credit Card	Ankit	₹ 800.00	Normal			Mouse	Mouse

9. Task 9 – Pivot Table & Charts

Objective

To summarize sales data using a Pivot Table and visualize the results using a chart.

Pivot Table

A Pivot Table was created using:

- **Rows:** Category
- **Values:** Sum of Sales_Value

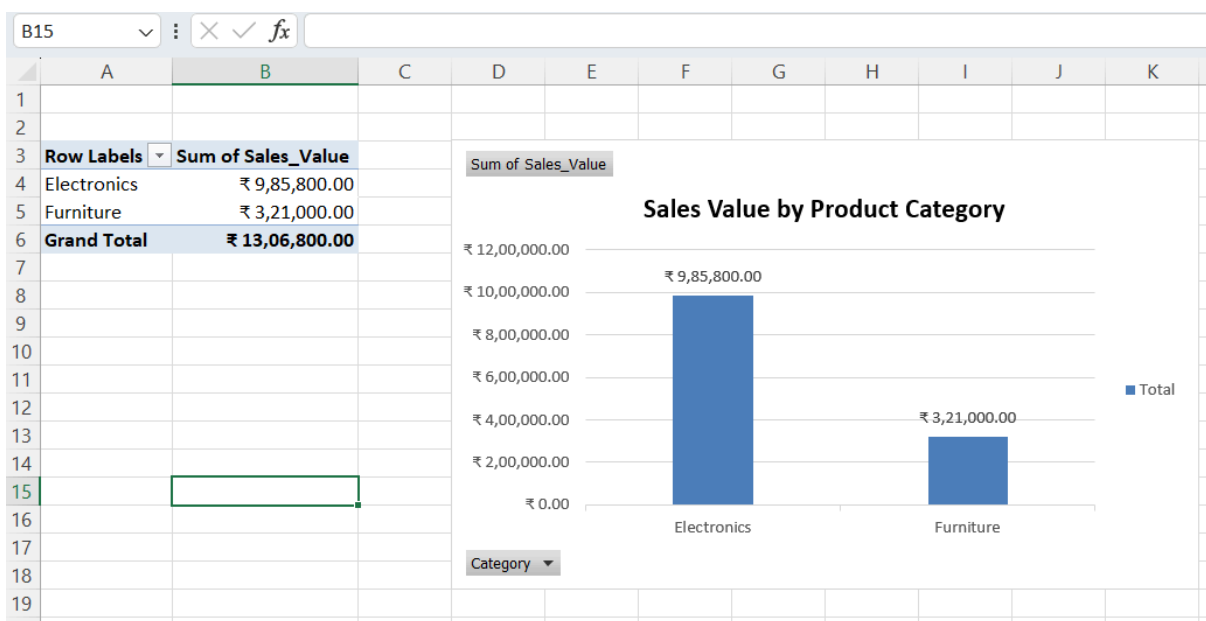
Chart

A column chart titled "**Sales Value by Product Category**" was created based on the Pivot Table.

Explanation

The chart compares the total sales value generated by each product category. **Electronics generated the highest sales value of ₹9,85,800**, while Furniture generated ₹3,21,000.

Screenshot



10. Google Drive Link:

Google Drive folder containing all files:

 week_3_assignment